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### Fan and light units and associated mounting arrangements for use at a loading dock

#### Abstract

Example fan and light units and associated mounting arrangements for use at a loading dock are disclosed. An example apparatus includes a frame including a bar to be pivotally coupled to a support system via a point of attachment on the bar. The apparatus includes a fan and a controller to be coupled to the frame with the bar extending in a first direction away from the controller and toward the point of attachment, the controller to control operation of the fan. The apparatus includes a handle to be connected to a portion of the frame extending away from the controller in a second direction opposite the first direction, the handle spaced apart from a housing of the fan, the handle to be lower than the controller and lower than at least a midpoint of the fan when the apparatus is supported by the support system.

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## References Cited

### U.S. PATENT DOCUMENTS

| Patent No. | Issued Date | Patentee Name   | U.S. Cl.  | CPC         |
|------------|-------------|-----------------|-----------|-------------|
| 1125346    | 12/1914     | Kramer          | N/A       | N/A         |
| 2538655    | 12/1950     | Preston         | N/A       | N/A         |
| 3275878    | 12/1965     | Wilbanks        | N/A       | N/A         |
| 3475844    | 12/1968     | Uda et al.      | N/A       | N/A         |
| 3782443    | 12/1973     | Clauss          | N/A       | N/A         |
| 4424006    | 12/1983     | Armbruster      | 417/313   | F04D 25/084 |
| 4651966    | 12/1986     | Suzuki          | N/A       | N/A         |
| 4696450    | 12/1986     | Huang           | N/A       | N/A         |
| 4697430    | 12/1986     | Toyoda et al.   | N/A       | N/A         |
| D296369    | 12/1987     | Harvey          | N/A       | N/A         |
| 4856968    | 12/1988     | Armbruster      | N/A       | N/A         |
| 4973012    | 12/1989     | Hashizume       | N/A       | N/A         |
| 4973016    | 12/1989     | Hertenstein     | 248/282.1 | F21V 21/108 |
| 5062029    | 12/1990     | Engel           | N/A       | N/A         |
| D327131    | 12/1991     | Vrancart et al. | N/A       | N/A         |
| 5148849    | 12/1991     | Faludy          | N/A       | N/A         |
| 5171086    | 12/1991     | Baloochi        | N/A       | N/A         |
| D340405    | 12/1992     | Anderson        | N/A       | N/A         |
| 5287253    | 12/1993     | Blorstad        | N/A       | N/A         |
| D359225    | 12/1994     | Reed            | N/A       | N/A         |
| 5575097    | 12/1995     | Chou et al.     | N/A       | N/A         |
| 5603562    | 12/1996     | Huang           | N/A       | N/A         |

|          |         |                  |         |              |
|----------|---------|------------------|---------|--------------|
| D378863  | 12/1996 | Wang             | N/A     | N/A          |
| 5667188  | 12/1996 | Bettinsoli       | N/A     | N/A          |
| 5709458  | 12/1997 | Metz             | N/A     | N/A          |
| 5738056  | 12/1997 | Mikame           | N/A     | N/A          |
| D397468  | 12/1997 | Zeller           | N/A     | N/A          |
| 5791763  | 12/1997 | Kam-Hoi          | N/A     | N/A          |
| 5996898  | 12/1998 | Parker et al.    | N/A     | N/A          |
| 6183203  | 12/2000 | Grintz           | 416/246 | F16M 11/2014 |
| D444260  | 12/2000 | Dolan            | N/A     | N/A          |
| 6244821  | 12/2000 | Boyd et al.      | N/A     | N/A          |
| D454975  | 12/2001 | De'Armond        | N/A     | N/A          |
| 6589016  | 12/2002 | Boyd et al.      | N/A     | N/A          |
| 6692229  | 12/2003 | Metz             | N/A     | N/A          |
| 6767117  | 12/2003 | De' Armond       | N/A     | N/A          |
| 6812849  | 12/2003 | Ancel            | N/A     | N/A          |
| 6817835  | 12/2003 | Boyd et al.      | N/A     | N/A          |
| 6859644  | 12/2004 | Wang             | N/A     | N/A          |
| 6890149  | 12/2004 | Metz et al.      | N/A     | N/A          |
| 7101064  | 12/2005 | Ancel            | N/A     | N/A          |
| 7147336  | 12/2005 | Chou             | N/A     | N/A          |
| D537196  | 12/2006 | Summerford       | N/A     | N/A          |
| D542943  | 12/2006 | Wang             | N/A     | N/A          |
| D546980  | 12/2006 | Lo               | N/A     | N/A          |
| D549854  | 12/2006 | Takahashi        | N/A     | N/A          |
| 7257706  | 12/2006 | Zucker           | N/A     | N/A          |
| D566300  | 12/2007 | Lo               | N/A     | N/A          |
| D582578  | 12/2007 | Tsai et al.      | N/A     | N/A          |
| D593223  | 12/2008 | Komar            | N/A     | N/A          |
| D599931  | 12/2008 | Van Klinken      | N/A     | N/A          |
| D608047  | 12/2009 | Scheper          | N/A     | N/A          |
| 7762688  | 12/2009 | Dixon et al.     | N/A     | N/A          |
| 7823239  | 12/2009 | Hochstein et al. | N/A     | N/A          |
| D636928  | 12/2010 | Lin              | N/A     | N/A          |
| D647646  | 12/2010 | Tsung            | N/A     | N/A          |
| 8069520  | 12/2010 | Mattucci         | N/A     | N/A          |
| D664272  | 12/2011 | Lee              | N/A     | N/A          |
| 8366301  | 12/2012 | Hudson et al.    | N/A     | N/A          |
| 8408761  | 12/2012 | Ng               | N/A     | N/A          |
| D683883  | 12/2012 | Yorio et al.     | N/A     | N/A          |
| 8528719  | 12/2012 | Huttner          | N/A     | N/A          |
| D693947  | 12/2012 | Wiegel et al.    | N/A     | N/A          |
| 8590689  | 12/2012 | Wahl             | N/A     | N/A          |
| D701640  | 12/2013 | Wiegel et al.    | N/A     | N/A          |
| D702397  | 12/2013 | Wiegel et al.    | N/A     | N/A          |
| 9255699  | 12/2015 | Wiegel et al.    | N/A     | N/A          |
| D762917  | 12/2015 | Nankil           | N/A     | N/A          |
| D797989  | 12/2016 | Herft            | N/A     | N/A          |
| D804704  | 12/2016 | Xia et al.       | N/A     | N/A          |
| D818517  | 12/2017 | Szarawarski      | N/A     | N/A          |
| 10145503 | 12/2017 | Feldman et al.   | N/A     | N/A          |

|              |         |                    |         |              |
|--------------|---------|--------------------|---------|--------------|
| D844882      | 12/2018 | Schuyler           | N/A     | N/A          |
| D851805      | 12/2018 | Chen               | N/A     | N/A          |
| D853630      | 12/2018 | Wilcox             | N/A     | N/A          |
| D882862      | 12/2019 | Antoni             | N/A     | N/A          |
| D899674      | 12/2019 | He                 | N/A     | N/A          |
| D905225      | 12/2019 | Hidalgo            | N/A     | N/A          |
| D907842      | 12/2020 | Mackin             | N/A     | N/A          |
| D933283      | 12/2020 | Wiegel             | N/A     | N/A          |
| 11215346     | 12/2021 | Wiegel             | N/A     | N/A          |
| 11739924     | 12/2022 | Wiegel et al.      | N/A     | N/A          |
| D1018958     | 12/2023 | Wiegel et al.      | N/A     | N/A          |
| 2002/0037098 | 12/2001 | Beaty et al.       | N/A     | N/A          |
| 2002/0045420 | 12/2001 | Taillon            | N/A     | N/A          |
| 2002/0144782 | 12/2001 | Okuno et al.       | N/A     | N/A          |
| 2003/0067767 | 12/2002 | Liu                | N/A     | N/A          |
| 2003/0133977 | 12/2002 | Mahashabde         | N/A     | N/A          |
| 2003/0151912 | 12/2002 | Ancel              | N/A     | N/A          |
| 2004/0114921 | 12/2003 | Braun et al.       | N/A     | N/A          |
| 2004/0131470 | 12/2003 | Boyd               | N/A     | N/A          |
| 2004/0160199 | 12/2003 | Morgan             | N/A     | N/A          |
| 2004/0254725 | 12/2003 | Douville           | N/A     | N/A          |
| 2005/0168999 | 12/2004 | Sommers et al.     | N/A     | N/A          |
| 2005/0230592 | 12/2004 | Tillinghast et al. | N/A     | N/A          |
| 2005/0260943 | 12/2004 | Snyder             | 454/118 | B60H 1/00014 |
| 2005/0269043 | 12/2004 | Welsh              | N/A     | N/A          |
| 2006/0048631 | 12/2005 | Jakes et al.       | N/A     | N/A          |
| 2006/0049615 | 12/2005 | Day                | N/A     | N/A          |
| 2006/0133097 | 12/2005 | Doric et al.       | N/A     | N/A          |
| 2006/0133907 | 12/2005 | Bullock            | N/A     | N/A          |
| 2006/0213672 | 12/2005 | Mohr               | N/A     | N/A          |
| 2007/0003421 | 12/2006 | Kato               | N/A     | N/A          |
| 2007/0057807 | 12/2006 | Walters            | N/A     | N/A          |
| 2007/0214586 | 12/2006 | Mattucci et al.    | N/A     | N/A          |
| 2007/0273290 | 12/2006 | Ashdown            | N/A     | N/A          |
| 2008/0010757 | 12/2007 | Hochstein et al.   | N/A     | N/A          |
| 2009/0116961 | 12/2008 | Todd, Jr.          | N/A     | N/A          |
| 2009/0147120 | 12/2008 | Kurane             | N/A     | N/A          |
| 2009/0287352 | 12/2008 | Geist              | N/A     | N/A          |
| 2010/0052549 | 12/2009 | Hudson et al.      | N/A     | N/A          |
| 2010/0075588 | 12/2009 | Haneline           | N/A     | N/A          |
| 2010/0146719 | 12/2009 | Swessel et al.     | N/A     | N/A          |
| 2010/0201267 | 12/2009 | Bourquin et al.    | N/A     | N/A          |
| 2010/0240418 | 12/2009 | Matoba             | N/A     | N/A          |
| 2010/0291858 | 12/2009 | Toy                | N/A     | N/A          |
| 2010/0301773 | 12/2009 | Chemel et al.      | N/A     | N/A          |
| 2011/0001957 | 12/2010 | Braune et al.      | N/A     | N/A          |
| 2011/0051437 | 12/2010 | Ng                 | N/A     | N/A          |
| 2011/0203059 | 12/2010 | Whitley et al.     | N/A     | N/A          |
| 2011/0304840 | 12/2010 | Hsu et al.         | N/A     | N/A          |
| 2012/0012424 | 12/2011 | Bowman             | N/A     | N/A          |

|              |         |                  |         |             |
|--------------|---------|------------------|---------|-------------|
| 2012/0055757 | 12/2011 | Wahl             | N/A     | N/A         |
| 2012/0055758 | 12/2011 | Huttner          | N/A     | N/A         |
| 2013/0173064 | 12/2012 | Fadell           | N/A     | N/A         |
| 2014/0112791 | 12/2013 | Abell            | 416/159 | F04D 25/084 |
| 2014/0225509 | 12/2013 | Wiegel           | N/A     | N/A         |
| 2015/0176834 | 12/2014 | Avedon           | N/A     | N/A         |
| 2016/0169498 | 12/2015 | Lei et al.       | N/A     | N/A         |
| 2017/0074503 | 12/2016 | Horng et al.     | N/A     | N/A         |
| 2017/0144645 | 12/2016 | Metz et al.      | N/A     | N/A         |
| 2017/0168531 | 12/2016 | Casparian et al. | N/A     | N/A         |
| 2017/0343222 | 12/2016 | Markert et al.   | N/A     | N/A         |
| 2018/0335040 | 12/2017 | Rouse            | N/A     | F04D 25/084 |
| 2019/0144218 | 12/2018 | Hoofard et al.   | N/A     | N/A         |
| 2019/0264698 | 12/2018 | Parel            | N/A     | N/A         |
| 2020/0085162 | 12/2019 | Yoo et al.       | N/A     | N/A         |
| 2020/0284411 | 12/2019 | Wiegel et al.    | N/A     | N/A         |
| 2022/0178528 | 12/2021 | Wiegel et al.    | N/A     | N/A         |

#### **FOREIGN PATENT DOCUMENTS**

| <b>Patent No.</b> | <b>Application Date</b> | <b>Country</b> | <b>CPC</b> |
|-------------------|-------------------------|----------------|------------|
| 2589238           | 12/2005                 | CA             | N/A        |
| 2772932           | 12/2010                 | CA             | N/A        |
| 1084834           | 12/1993                 | CN             | N/A        |
| 1175661           | 12/1997                 | CN             | N/A        |
| 1382564           | 12/2001                 | CN             | N/A        |
| 101384205         | 12/2008                 | CN             | N/A        |
| 201313754         | 12/2008                 | CN             | N/A        |
| 102026781         | 12/2010                 | CN             | N/A        |
| 102398762         | 12/2011                 | CN             | N/A        |
| 102625890         | 12/2011                 | CN             | N/A        |
| 4031471           | 12/1991                 | DE             | N/A        |
| 69703670          | 12/2000                 | DE             | N/A        |
| 0799977           | 12/1999                 | EP             | N/A        |
| 1465563           | 12/2003                 | EP             | N/A        |
| 1983879           | 12/2007                 | EP             | N/A        |
| 2019252           | 12/2008                 | EP             | N/A        |
| 1769187           | 12/2008                 | EP             | N/A        |
| 2359731           | 12/2010                 | EP             | N/A        |
| 2428470           | 12/2011                 | EP             | N/A        |
| 2473779           | 12/2011                 | EP             | N/A        |
| 2428471           | 12/2012                 | EP             | N/A        |
| 437647            | 12/1911                 | FR             | N/A        |
| 869468            | 12/1941                 | FR             | N/A        |
| 983013            | 12/1964                 | GB             | N/A        |
| 2010229           | 12/1978                 | GB             | N/A        |
| 2123132           | 12/1983                 | GB             | N/A        |
| 1003642           | 12/2000                 | GR             | N/A        |
| 2000042970        | 12/1999                 | JP             | N/A        |
| 3211713           | 12/2000                 | JP             | N/A        |
| 2011026887        | 12/2010                 | JP             | N/A        |

|            |         |    |     |
|------------|---------|----|-----|
| 2011149300 | 12/2010 | JP | N/A |
| 2013504164 | 12/2012 | JP | N/A |
| 2012002624 | 12/2011 | MX | N/A |
| 2009003279 | 12/2008 | WO | N/A |
| 2009087261 | 12/2008 | WO | N/A |
| 2014123733 | 12/2013 | WO | N/A |
| 2015142871 | 12/2014 | WO | N/A |

## OTHER PUBLICATIONS

Rite-Hite, "CoolMan 2500 Trailer Fan Owner's and Installation Manual," Mar. 2004, 12 pages. cited by applicant

Mexican Institute of Industrial Property, "Office Action," issued in connection with Mexican Design Patent Application No. MX/f/2019/002484, dated Sep. 3, 2020, 7 pages. cited by applicant  
IP Australia, "Examination Report No. 1," issued in connection with Australian Patent Application No. 2020200426, dated Oct. 1, 2020, 8 pages. cited by applicant

United States Patent and Trademark Office, "Office Non-Final Action" issued in connection with U.S. Appl. No. 16/747,360, mailed on Oct. 21, 2020, 23 pages. cited by applicant

Mexican Institute of Industrial Property, "Office Action," issued in connection with Mexican Design Patent Application No. MX/a/2020/000844, dated Nov. 9, 2020, 5 pages. cited by applicant

Canadian Intellectual Property Office, "Examination Report," issued in connection with Canadian Design Patent Application No. 189648, dated Nov. 20, 2020, 2 pages. cited by applicant

Canadian Intellectual Property Office, "Office Action," issued in connection with Canadian Patent Application No. 3069222, dated Mar. 1, 2021, 5 pages. cited by applicant

Mexican Institute of Industrial Property, "Notice of Allowance," issued in connection with Mexican Design Patent Application No. MX/f/2019/002484, dated Apr. 26, 2021, 4 pages. cited by applicant

United States Patent and Trademark Office, "Notice of Allowance and Fee(s) Due," issued in connection with U.S. Appl. No. 29/703,602, mailed on May 12, 2021, 7 pages. cited by applicant

United States Patent and Trademark Office, "Final Office Action" issued in connection with U.S. Appl. No. 16/747,360, mailed on May 24, 2021, 20 pages. cited by applicant

IP Australia, "Examination Report No. 2," issued in connection with Australian Patent Application No. 2020200426, dated Jul. 15, 2021, 3 pages. cited by applicant

Mexican Institute of Industrial Property, "Notice of Allowance," issued in connection with Mexican Design Patent Application No. MX/f/2020/003192, dated Aug. 4, 2021, 5 pages. cited by applicant

United States Patent and Trademark Office, "Advisory Action," issued in connection with U.S. Appl. No. 16/747,360, mailed on Aug. 6, 2021, 4 pages. cited by applicant

United States Patent and Trademark Office, "Notice of Allowance and Fee(s) Due," issued in connection with U.S. Appl. No. 16/747,360, mailed on Sep. 7, 2021, 19 pages. cited by applicant

IP Australia, "Notice of Acceptance for Patent Application," issued in connection with Australian Application No. 2020200426, dated Sep. 13, 2021, 3 pages. cited by applicant

United States Patent and Trademark Office, "Notice of Allowability for a Design Application," issued in connection with U.S. Appl. No. 29/703,602, dated Sep. 15, 2021, 2 pages. cited by applicant

Canadian Intellectual Property Office, "Office Action," issued in connection with Canadian Patent Application No. 3,069,222, dated Oct. 6, 2021, 4 pages. cited by applicant

Canadian Intellectual Property Office, "Office Action," issued in connection with Canadian Patent Application No. 201546, dated Oct. 19, 2021, 1 page. cited by applicant

Canadian Intellectual Property Office, "Examination Report," issued in connection with Canadian Patent Application No. 201548, dated Oct. 19, 2021, 1 page. cited by applicant

Canadian Intellectual Property Office, "Examination Report," issued in connection with Canadian Patent Application No. 201545, dated Oct. 19, 2021, 1 page. cited by applicant

Canadian Intellectual Property Office, "Examination Report," issued in connection with Canadian Patent Application No. 201547, dated Oct. 19, 2021, 1 page. cited by applicant

Canadian Intellectual Property Office, "Office Action," issued in connection with Canadian Patent Application No. 201549, dated Oct. 19, 2021, 1 page. cited by applicant

United States Patent and Trademark Office, "Supplemental Notice of Allowability," issued in connection with U.S. Appl. No. 16/747,360, mailed on Nov. 8, 2021, 3 pages. cited by applicant

Canadian Intellectual Property Office, "Notice of Registration," issued in connection with Canadian Patent Application No. 201546, dated Jan. 28, 2022, 8 pages. cited by applicant

Canadian Intellectual Property Office, "Notice of Registration," issued in connection with Canadian Patent Application No. 201545, dated Jan. 28, 2022, 8 pages. cited by applicant

Canadian Intellectual Property Office, "Notice of Registration," issued in connection with Canadian Patent Application No. 201547, dated Jan. 28, 2022, 8 pages. cited by applicant

Canadian Intellectual Property Office, "Notice of Registration," issued in connection with Canadian Patent Application No. 201548, dated Jan. 28, 2022, 8 pages. cited by applicant

Canadian Intellectual Property Office, "Notice of Registration," issued in connection with Canadian Patent Application No. 201549, dated Jan. 28, 2022, 8 pages. cited by applicant

Canadian Intellectual Property Office, "Office Action," issued in connection with Canadian Patent Application No. 3,069,222 mailed on May 5, 2022, 3 pages. cited by applicant

United States Patent and Trademark Office, "Non-Final Office Action", issued in connection with U.S. Appl. No. 17/567,615, Jul. 15, 2022, 11 pages. cited by applicant

Australian Government, IP Australia, "Examination Report No. 1," issued in connection with Australian Patent Application No. 2021290321, dated Dec. 12, 2022, 5 pages. cited by applicant

United States Patent and Trademark Office, "Final Office Action", issued in connection with U.S. Appl. No. 17/567,615, Jan. 17, 2023, 14 pages. cited by applicant

Canadian Intellectual Property Office, "Office Action," issued in connection with Canadian Patent Application No. 3,069,222, mailed on Feb. 27, 2023, (5 pages). cited by applicant

United States Patent and Trademark Office, "Notice of Allowance and Fee(s) Due," issued in connection with U.S. Appl. No. 17/567,615, mailed on Mar. 29, 2023, 9 pages. cited by applicant

Mexican Intellectual Property Office, "Office Action," issued in connection with Mexican Design Patent Application No. MX/a/2020/000844, dated Mar. 30, 2023, 6 pages. cited by applicant

IP Australia, "Notice of Acceptance for Patent Application," issued in connection with Australian Patent Application No. 2021290321, dated Apr. 5, 2023, 3 pages. cited by applicant

United States Patent and Trademark Office, "Non-Final Office Action," issued in connection with U.S. Appl. No. 29/805,971, mailed on Jun. 9, 2023, 7 pages. cited by applicant

United States Patent and Trademark Office, "Supplemental Notice of Allowability," issued in connection with U.S. Appl. No. 17/734,792, mailed on Jun. 14, 2023, 2 pages. cited by applicant

United States Patent and Trademark Office, "Notice of Allowability," issued in connection with U.S. Appl. No. 17/567,615, mailed on Jul. 5, 2023, 2 pages. cited by applicant

Australian Government, IP Australia, "Notice to Grant," mailed in connection with Australian Patent Application No. 2021290321, mailed on Aug. 3, 2023, 1 page. cited by applicant

Mexican Institute of Industrial Property, "Notice of Allowance," issued in connection with Mexican Patent Application No. MX/a/2020/000844, dated Sep. 6, 2023, 6 pages. cited by applicant

United States Patent and Trademark Office, "Notice of Allowance and Fee(s) Due," issued in connection with U.S. Appl. No. 29/805,971, issued on Oct. 27, 2023, 8 pages. cited by applicant

Canadian Patent Office, "Conditional Notice of Allowance," issued in connection with Canadian Patent Application No. 3,069,222, mailed on Mar. 27, 2024, 3 pages. cited by applicant

IP Australia, "Examination report No. 1," issued in connection with Australian Patent Application No. 2023206105, issued on Jun. 7, 2024, 7 pages. cited by applicant

Australian Government, IP Australia, "Examination report No. 2," issued in connection with AU Application No. 2023206105, issued on Nov. 25, 2024, 2 pages. cited by applicant

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## **Background/Summary**

RELATED APPLICATIONS (1) This patent arises from a continuation of U.S. patent application Ser. No. 17/567,615, which was filed on Jan. 3, 2022, which was a continuation of U.S. patent application Ser. No. 16/747,360 (now U.S. Pat. No. 11,215,346), which was filed on Jan. 20, 2020, which was a continuation-in-part of U.S. Design patent application No. 29/703,602 (now U.S. Design Pat. No. D933,283), which was filed on Aug. 28, 2019, and which claims priority to U.S. Provisional Patent Application No. 62/816,050, which was filed on Mar. 8, 2019. U.S. patent application Ser. No. 17/567,615, U.S. patent application Ser. No. 16/747,360, U.S. Provisional Patent Application No. 62/816,050, and U.S. Design patent application No. 29/703,602 are incorporated herein by reference in their entireties. Priority to U.S. patent application Ser. No. 17/567,615, U.S. patent application Ser. No. 16/747,360, U.S. Provisional Patent Application No. 62/816,050, and U.S. Design patent application No. 29/703,602 is claimed.

### **FIELD OF THE DISCLOSURE**

(1) This patent generally pertains to loading dock equipment and more specifically to fan and light units and associated mounting arrangements for use at a loading dock.

### **BACKGROUND**

(2) A typical loading dock of a building includes an elevated platform for transferring cargo to and from a vehicle, such as a truck or trailer. A passageway defined in the exterior building wall is generally positioned above and adjacent to the elevated platform, and may be selectively blocked by a door to separate the interior and exterior environments or unblocked to enable transfer of a load between the building and the vehicle therethrough. Loading docks include various pieces of equipment to facilitate the loading and unloading operations. Examples of such equipment include dock levelers, vehicle restraints, weather barriers, lights for illuminating the vehicle's cargo bay, and fans for ventilating the cargo bay.

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## **Description**

### **BRIEF DESCRIPTION OF THE DRAWINGS**

(1) FIG. 1 is a side view of a vehicle at a loading dock with an example mounted fan and light apparatus constructed in accordance with teachings disclosed herein.

(2) FIG. 2 is a perspective view of the example mounted fan and light apparatus shown in FIG. 1 from inside the building.

(3) FIG. 3 is another perspective view of the example mounted fan and light apparatus shown in FIG. 1 from inside the building.

(4) FIG. 4 is another perspective view of the example mounted fan and light apparatus shown in FIG. 1.

(5) FIG. 5 is another perspective view of the example mounted fan and light apparatus shown in FIG. 1.



- (6) FIG. 6 is another perspective view of the example mounted fan and light apparatus shown in FIG. 1.
- (7) FIG. 7 is another perspective view of the example mounted fan and light apparatus shown in FIG. 6.
- (8) FIG. 8 is a top view of the example mounted fan and light apparatus shown in FIG. 7.
- (9) FIG. 9 is a front view of the example mounted fan and light apparatus shown in FIG. 8.
- (10) FIG. 10 is a back view of the example mounted fan and light apparatus shown in FIG. 9.
- (11) FIG. 11 is a perspective view of an example fan and light unit constructed in accordance with teachings disclosed herein.
- (12) FIG. 12 is a perspective view of another example mounted fan and light apparatus constructed in accordance with teachings disclosed herein.
- (13) FIG. 13 is another perspective view of the example mounted fan and light apparatus shown in FIG. 12.
- (14) FIG. 14 is a perspective view of an example vertically adjustable mounting bracket constructed in accordance with teachings disclosed herein.
- (15) FIG. 15 is a diagram showing an example mounting arrangement of the example mounted fan and light apparatus at a loading dock in accordance with teachings disclosed herein.
- (16) FIG. 16 is a diagram showing an example user interface on an example controller of an example fan and light unit constructed in accordance with teachings disclosed herein.
- (17) FIG. 17 is a perspective view of the example mounted fan and light apparatus shown in FIG. 9 but with the fan housing turned ninety degrees.
- (18) FIG. 18 is a perspective view of the example mounted fan and light apparatus shown in FIG. 12 but with the fan housing turned ninety degrees.
- (19) FIG. 19 is a perspective view of another example mounted fan and light apparatus constructed in accordance with teachings disclosed herein.
- (20) FIG. 20 is a perspective view of another mounted fan and light apparatus constructed in accordance with teachings disclosed herein.
- (21) FIG. 21 is a perspective view of an example frame of the example fan and light unit with the controller attached.
- (22) FIG. 22 is a perspective view of another example frame of the example fan and light unit with the controller attached.
- (23) FIG. 23 is a perspective view of another example frame of the example fan and light unit with the controller attached.
- (24) FIG. 24 is a perspective view of another example frame of the example fan and light unit with the controller attached.
- (25) FIG. 25 is a perspective view of another example frame of the example fan and light unit with the controller attached.
- (26) FIG. 26 is a perspective view of another example frame of the example fan and light unit with the controller attached.

#### DETAILED DESCRIPTION

(27) Example mounted light and fan units disclosed herein for use vehicle loading docks include both a light and a fan. While cargo is being transferred to or from the cargo bay of a vehicle parked at the dock, the light illuminates the cargo bay while the fan ventilates it. In some examples, a fan axis of the fan's impeller can be changed selectively to horizontal and vertical orientations. Some examples of the mounted light and fan include an articulated arm mounted to a building wall by way of a bracket that is vertically adjustable. In some examples, a curved frame couples the articulated arm to the fan's housing, wherein the frame serves as a conduit for electrical wiring extending between the light and a controller. Some examples of the mounted light and fan include dual-grip handlebars for aiming the fan and the light.

(28) FIGS. 1-3 show an example mounted light and fan apparatus 10 secured to a building structure

**12** (e.g., a wall, a doorframe, a bollard, a post, etc.) at a vehicle loading dock **14**. The example apparatus **10** includes a fan **16** for ventilating an interior cargo bay **20** of a vehicle **22** (e.g., a truck, a trailer, etc.) parked at dock **14**. Further, the example apparatus **10** includes a light **18** for illuminating the interior cargo bay **20** of the vehicle **22**.

(29) In the illustrated example, the dock **14** includes a building **24**, a wall **26** with a doorway **28**, a driveway or approach **30** leading to the doorway **28**, a door **32**, an elevated floor **34** of the building **24**, a dock leveler **36** having an adjustable deck **38** and an extendible lip **40**, and a vehicle restraint **42**. Opening the door **32** of the dock **14** and a rear door **44** of the vehicle **22** enables cargo to be transferred between the building **24** and the vehicle's cargo bay **20**. To aid loading and unloading operations, the dock leveler **36** provides a bridge between the dock floor **34** and the vehicle **22** across which the cargo can be transferred between the vehicle **22** and the building **24** while the vehicle restraint **42** helps secure the vehicle **22** at the dock **14**. During the loading and/or unloading process(es), the mounted light and fan apparatus **10** helps illuminate and ventilate the vehicle's cargo bay **20** while reducing (e.g., minimizing) obstruction to movement through the doorway **28**.

(30) Details and alternatives of the mounted light and fan apparatus **10** are shown in FIGS. **4-20**. Some examples of the apparatus **10** comprise a mounting bracket **46**, a fan and light unit **48** (FIG. **11**), and an articulated arm **50**. The mounting bracket **46** is secured to a building structure **12**. The fan and light unit **48** comprises a fan **16**, a light **18**, and a frame **52** coupled to both the fan **16** and the light **18**. The fan and light unit **48** is also referred to herein as an illuminating fan assembly. The articulated arm **50** connects the fan and light unit **48** to the mounting bracket **46** to provide an adjustable support means for selectively positioning and aiming the fan and light unit **48**. In some examples, the fan **16** and the light **18** are attached to the frame **52** to point in substantially the same direction (e.g., within 5 degrees of one another). In other examples, the light **18** may point in a direction that is offset relative to the direction in which the fan **16** is pointing. In some examples, the relatively direction of the fan **16** and the light **18** may be user-adjustable. In other examples, both the fan **16** and the light **18** are rigidly affixed to the frame so that the relative direction of the fan **16** and the light **18** is fixed. That is, in some examples, the fan **16** and the light **18** may be held in fixed relationship relative to one another.

(31) Referring to FIG. **11**, some examples of the fan and light unit **48** comprise the fan **16**, the light **18**, the frame **52**, and one or more controllers **54** for the light **18** and/or the fan **16**, a first handle **56** and a second handle **58**. In some examples, the first handle **56** comprises a portion of or extends from the frame **52**. In some examples, the second handle **58** comprises a portion of or extends from the housing of the fan **16**. FIG. **16** shows an example user interface **60** of the controller **54**. In this example, the user interface **60** is specific to operation of the light **18**. Separate controls for operation of the fan **16** (e.g., FIG. **7, 9, 17, 19**) may be provided via another user interface of the controller **54**. In other examples, controls for the fan **16** may be provided via a user interface of a separate controller specific to the fan. Alternatively, the controller **54** may be used to operate either or both of the light **18** and the fan **16** to selectively power each component (e.g., the fan **16** and/or the light **18**) on or off, increase/decrease power/intensity, enable automatic (sensor-controlled) output modulation, and/or set/start/stop a timer for changing a power state for one or both of the components. In some examples, the automatic output modulation enables the fan **16** and/or the light **18** to be controlled by one or more sensors (e.g., motion, ambient light, temperature, etc.). In some examples, the one or more sensors may be incorporated and/or integrated with the fan **16**, the light **18**, and/or the controller **54**. Additionally or alternatively, the one or more sensors may be separate from but carried by the fan **16**, the light **18**, the controller **54**, and/or the frame **52**. Further, in some examples, one or more of the sensors may be spaced apart from the fan and light unit **48** but in communication with the fan **16**, the light **18**, and/or the controller **54**. In some examples, the controller **54** powers off the fan **16** and/or the light **18** after a certain time period in which motion in the area proximate the unit **48** was not detected by a motion sensor. In other examples, the controller **54** powers on the fan **16** and/or the light **18** when motion in the area proximate the unit

**48** is detected by a motion sensor. In some examples, the motion sensor is positioned to monitor the area toward which the fan **16** and the light **18** are pointed. Additionally or alternatively, the motion sensor is positioned to monitor an area other than in the direction that the fan **16** and the light **18** are pointing. Signals from various types of sensors could be used alone or in combination to control the fan **16** and/or the light **18** in different use cases and environmental scenarios.

(32) In some examples, the frame **52** comprises a hollow bar or tubular conduit **62** with an internal channel through which an electrical wire **64** (represented by the dashed line in FIG. **11**) runs between the controller **54** and the light **18**. In some examples, the bar or conduit **62** is rigid to provide structural support for the other components of the fan and light unit **48**. In the illustrated example, the conduit **62** has a generally square cross-section. However, other cross-sectional shapes (e.g., circular, rectangular, etc.) may alternatively be used. The electrical wire **64** is just one example of a controlling communication link **66** operatively connecting the controller **54** to the light **18** and/or the fan **16**. In some examples, the tubular conduit **62** curves and/or bends about a fan axis **68** and, more generally, about a housing **72** of the fan to reduce (e.g., minimize) structural weight, enhance the unit's appearance, and/or to make it easier to feed the wire **64** through the conduit **62** during assembly. More particularly, in some examples, the frame **52** extends between the controller **54** and the light **18** with a bend such that the controller **54** is positioned proximate a backside of the fan **16** with the light **18** being positioned proximate (but spaced apart from) a top and/or side of the fan **16**. In some examples, all three of the fan **16**, the light **18**, and the controller **54** are aligned within a common plane defined by the curve or bend in the conduit **62**. Further, in some examples, the relative position of the fan **16**, the light **18** and the controller **54** (due to the shape of the frame **52**) generally corresponds to a right triangle with the fan **16** at the right angle of the triangle. In some examples, to enable this arrangement, the tubular conduit **62** curves approximately (e.g., within 5 degrees of) 90 degrees. In other examples, the frame **52** may be formed of straight pieces of conduit **62** connected at joints defining suitable angles (e.g., two pieces connected by a 90 degree joint, three pieces connected in series by two 135 degree joints, etc.). In some examples, additional wiring **70** for the fan **16** and/or the light **18** is strung external of the frame **52** and the articulated arm **50**, as shown in FIG. **5**. In some such examples, the controller **54** may be spaced apart from the fan and light unit **48**.

(33) In the illustrated examples, the fan **16** comprises a fan housing **72** containing a centrifugal impeller **74** (FIG. **1**). The fan housing **72** defines a suction air inlet **76** and an air discharge outlet **78**. An electric motor inside housing **72** rotates the impeller **74** about a fan axis **68** to force air **80** from the inlet **76**, into the housing **72**, out through the outlet **78** toward the direction the fan **16** is pointing (e.g., into the vehicle's cargo bay **20**). The articulated arm **50** allows the fan **16** and the light **18** to be positioned and repositioned with respect to the structure to which it is mounted.

(34) In some examples, the articulated arm **50** comprises a proximal arm **82** pivotally coupled to a distal arm **84**. The proximal arm **82** is coupled to the mounting bracket **46**, and the distal arm **84** is coupled to the fan and light unit **48** via the frame **52**. The articulation of arm **50** is accomplished by way of several pivotal connections. In some examples such as that shown in FIG. **6**, multiple ones of the pivotal axes of these connections are generally parallel (e.g., parallel to within 5 degrees) to one another. For example, a first pivotal connection **86**, defining a first axis **88** (generally vertical in the illustrated example), couples the proximal arm **82** to the mounting bracket **46** such that proximal arm **82** is pivotal about the first axis **88** relative to the mounting bracket **46**. A second pivotal connection **90**, defining a second axis **92** (generally vertical in the illustrated example), couples the distal arm **84** to the proximal arm **82** such that the distal arm **84** is pivotal relative to the proximal arm **82** about the second axis **92**. A third pivotal connection **94**, defining a third axis **96** (generally vertical in the illustrated example), couples a positioning bracket **97** to the distal arm **84** such that positioning bracket **97** is pivotal relative to the distal arm **84** about the third axis **96**. As shown in the illustrated example, the proximal arm **82** and the distal arm **84** include elongate beams that connected at their ends. Therefore, the proximal arm **82** and the distal arm **84** extend in

elongate directions that are substantially perpendicular to the first, second, and third axes **88, 92, 96**. A fourth pivotal connection **98** defining a fourth axis **100** (generally horizontal in the illustrated example) couples the fan and light unit **48** via the frame **52** to the positioning bracket **97** along the third axis **96** below the third pivotal connection **94**. Thus, in the illustrated example, the fourth axis **100** is substantially (e.g., within 5 degrees) perpendicular to the first, second, and third axes **88, 92, 96**. As a result, the fan and light unit **48** (including the frame **52**) is independently positional pivotally about the third axis **96** and the fourth axis **100**, the third and fourth axes **96, 100** intersecting. In some examples, the tightness of the bolts and/or other hardware at the respective pivotal connections **86, 90, 94, 98** controls the freedom with which the components of the articulated arm **50** rotate relative to one another. That is, when the bolts and/or other hardware is relatively loose, the components may easily rotate relative to one another. By contrast, when the bolts are tightened, rotation at the pivotal connections may be significantly reduced and substantially prevented (at least by manual force).

(35) Although the actual construction details of the articulated arm **50** may vary, some examples of the proximal arm **82** comprises an upper beam **102** and a lower beam **104**, wherein the distal arm **84** is interposed between the upper and lower beams **102, 104** at the second pivotal connection **90**, and both the upper and lower beams **102, 104** are coupled to the mounting bracket **46** via the first pivotal connection **86**. In some examples, one or more spacers **105** are positioned at intermediate position(s) along the proximal arm **82** between the upper and lower beams **102** to provide support the proximal arm **82**. To provide the distal arm **84** with additional support, some examples of the articulated arm **50** include a diagonal tensile brace **106** having one end **108** connected to the proximal arm **82** and an opposite end **108** connected to the distal arm **84**. The illustrated example shows the diagonal tensile brace **106** having a central portion **110** that lies at an incline, i.e., the central portion **110** is displaced out of parallel alignment with both the proximal arm **82** and the distal arm **84**.

(36) In some examples, the distal arm **84** comprises a base segment **112** and an extension segment **114**. In some examples, the base segment **112** is coupled to the proximal arm **82** via the second pivotal connection **90**, the extension segment **114** is coupled to the frame **52** via the third pivotal connection **94** and the fourth pivotal connection **98**, and the extension segment **114** is telescopically coupled to the base segment **112**. The telescopic connection makes it possible to readily adjust the length of the distal arm **84**. In some examples, the particular length at which the extension **114** extends out from the base segment **112** is fixed via a bolt **115** extending through both the base segment **112** and the extension **114** at the distal end **108** of the tensile brace **106**. In some examples, the extension **114** includes a series of spaced apart holes distributed along its length through which the bolt **115** may extend to maintain the extension **114** at a fixed telescopic position relative to the base segment **112**.

(37) To reduce (e.g., minimize) obstruction by the unit **48** of traffic between the vehicle cargo bay **20**, through the doorway **28**, and into the building **24** (i.e., the traffic way), and/or to reduce (e.g., minimize) obstruction by traffic (i.e., forklifts, loads, dock workers) of the air stream(s) and light beam(s) produced by the unit **48**, the fan and light unit **48** may be positioned near the upper corner of the doorway **28**. With the articulated arm **50** (including positioning bracket **97**) providing various degrees of adjustable freedom and length, a dockworker may use the first handle **56** to easily position the fan and light unit **48** relative to the building structure **12**, the doorway **28**, the vehicle **22**, and/or traffic way associated with the dock between various use positions and various stored positions. In some examples, the first handle **56** may also be used to aim the fan **16**, and associated air stream(s), and the light **18**, and associated light beam(s), relative to the building structure **12**, the doorway **28**, the vehicle **22**, and/or traffic way associated with the dock. To increase (e.g., maximize) the elevation of the fan **16** and the light **18** with respect to the dock floor **34** while keeping the unit **48** within a dockworker's reach, the first handle **56** may be positioned or located at or near the lower edge of the unit **48** (i.e., lower than the fan axis **68** (e.g., a midpoint of

the fan **16**), the controller **54**, and/or the light **18**). In some examples, the elevation of the controller **54** is positioned below that of the light **18** to increase (e.g., maximize) the elevation of the light **18** in order to reduce (e.g., minimize) obstruction of the light shining into the cargo bay **20** while keeping the controller **54** within reach of the dockworker. The first handle **56** may be of a shape that enables ergonomic positioning of the unit **48**. In the illustrated example, the first handle **56** is in the form of a handlebar with two handgrips **116** (generally perpendicular to the third axis **96** and generally parallel to the fourth axis **100**) to make it easier for the dockworker to position and aim the fan **16** and the light **18**. In some examples, the first handle **56** is an integral feature of the frame **52**. Some examples of the fan and light unit **48** include the second handle **58** with a handgrip generally perpendicular to those of the first handle **56**, which provides a dockworker with more options for gripping to position and aim the fan and light unit **48**. In some examples, the second handle **58** is an integral feature of the fan housing **72**. In operation, a dockworker can grasp some combination of the handgrips **116** and the second handle **58** simultaneously (e.g., one hand on the first handle **56** and the other hand on second handle **58**) to position the fan and apply a rotational moment about the axis **96** to aim the fan **16** and the light **18** horizontally (e.g., when the third axis **96** is oriented vertically) and about the axis **100** to aim the fan **16** and the light **18** vertically (e.g., when the fourth axis **100** is oriented horizontally).

(38) Referring to FIGS. **12-15**, some examples of the apparatus **10** have an adjustable mounting bracket **46'** comprising a track **118** and a track follower **120** that render the articulated arm **50** and the fan and light unit **48** movable along the track **118** relative to the structure **12** to which the bracket **46'** is mounted. In some examples, the track extends in a direction that is substantially parallel (e.g., within 5 degrees) to the first axis **88** and, therefore, substantially perpendicular (e.g., within 5 degrees) to the direction in which the proximal arm **82** extends from the bracket **46'**. As a result, when the adjustable mounting bracket **46'** is mounted to the structure **12** with the track **118** extending vertically, the articulated arm **50** extends horizontally with the fan and light unit **48** being enabled to be selectively raised (vertically) for additional clearance underneath the unit **48** or lowered to an elevation more suitable for positioning and/or ventilating and/or illuminating the vehicle's cargo bay **20** and then being maintained in that vertical position until being subsequently and selectively raised or lowered. In some examples, the wall mounting bracket **46'** is adjustable by way of an actuator **122** such as, for example, a crank-up screw jack **124** manually powered by a crank **126**, a motorized linear actuator, a powered cylinder, a spring and/or various combinations thereof, etc.

(39) Some examples of the apparatus **10** include a fan and light unit **48** with a different orientation of the fan **16** with respect to the light **18** and/or the frame **52**. For example, the fan **16** can be set at a horizontal orientation with the fan axis **68** being substantially horizontal with the discharge outlet **78** positioned so as to be at the top of fan **16**, as shown in FIGS. **6-10**, or a vertical orientation with the fan axis **68** being substantially vertical with the discharge outlet **78** positioned so as to be to the side of fan **16**, as shown in FIGS. **17-20**. To achieve this, some examples of the apparatus **10** have a frame plate **130** (FIG. **11**) on the frame **52** and a mating mounting surface **128** on the fan housing **72**. In some examples, the controller **54** is positioned adjacent and/or attached to the frame plate **130** on an opposite side to the fan **16**. Bolt hole patterns **131** in the frame plate **130** and corresponding holes in the mounting surface **128** are symmetrical such that the frame plate **130** can be attached to the mounting surface **128** selectively in a first orientation (e.g., FIGS. **6-10**) and a second orientation (FIGS. **17-20**). In some examples, the pattern of holes **131** is positioned so as to surround the controller **54** that is also attached to the frame plate **130**. As most easily seen in FIG. **17**, the alternate mounting arrangements for the unit **48** enables the unit **48** to be positioned relative to the building structure **12**, the doorway **28**, the vehicle **22**, and/or traffic way associated with the dock such that a reduced (e.g., minimal) portion of the apparatus **10** (particularly, the outlet **78** of the fan **16**) extends into or near the traffic way while still ventilating and illuminating the cargo bay **20**. That is, as shown in the illustrated examples, the outlet **78** of the fan **16** extends across

substantially the full width of the fan **16** when oriented upright (e.g., as shown most clearly in FIG. **9**) such that the entire fan housing **72** would need to be within the doorway **28** and/or traffic way to enable the entire outlet **78** to be exposed to the cargo bay **20** of the vehicle **22**. By contrast, when the fan **16** is oriented on its side with the outlet **78** oriented vertically to one side (e.g., as shown in FIG. **17**) a considerable portion of the fan housing **72** may be positioned outside of the doorway **28** and/or the traffic way if the light **18** is positioned directly adjacent the edge of the doorway **28** (e.g., just inside the door jamb) while still enabling the entire outlet **78** to be exposed to the cargo bay **20** of the vehicle **22**.

(40) In the examples shown in FIGS. **19** and **20**, a vertical single handgrip handle **55** extends down from the frame **52** (perpendicular to the second handle **58** of the 90 degree rotated fan) replacing the horizontal handlebar-style handle **56**. In operation, a dockworker can grasp the handles **55**, **58** simultaneously (e.g., one hand on the single handgrip handle **55** and the other hand on the second handle **58**) to position the fan **16** and apply a rotational moment about the axis **96** to aim the fan **16** and the light **18** horizontally (e.g., when the third axis **96** is oriented vertically) and about the axis **100** to aim the fan **16** and the light **18** vertically (e.g., when the fourth axis **100** is oriented horizontally).

(41) In addition to being able to mount the fan **16** horizontally or vertically, it should be noted that in some examples, such as the examples shown in FIGS. **19** and **20**, the fan **16** can be flipped over 180 degrees about a horizontal axis. FIG. **20**, for instance, shows the fan **16** being mounted 180-degrees from its orientation shown in FIG. **19**. Being able to mount the fan **16** in either position can be advantageous depending on whether the mounting bracket **46** or **46'** is mounted on the right or left side of the doorway **28**.

(42) As mentioned above, the hollow bar or tubular conduit **62** of the frame **52** may have different cross-sectional shapes and/or have different types of bends along its length between the controller **54** and the light **18**. Several different examples are shown in FIGS. **21-26**. In particular, FIG. **21** illustrates the frame **52** with the example conduit **62** having a square cross-section and a curved bend. FIG. **22** illustrates the frame **52** with the example conduit **62** having a round cross-section and a curved bend. FIG. **23** illustrates the frame **52** with the example conduit **62** having a square cross-section and an approximately 90 degree bend. FIG. **24** illustrates the frame **52** with the example conduit **62** having a round cross-section and an approximately 90 degree bend. FIG. **25** illustrates the frame **52** with the example conduit **62** having a square cross-section and two approximately 135 degree bends. FIG. **26** illustrates the frame **52** with the example conduit **62** having a round cross-section and two approximately 135 degree bends.

(43) Example 1 includes an apparatus comprising a light, a fan, and a frame to support the light and the fan in fixed relationship to one another, the frame to pivotally couple to an articulated arm, the articulated arm to be coupled to a building structure adjacent a vehicle loading dock to enable selective positioning of the light and the fan relative to a doorway at the loading dock

(44) Example 2 includes the apparatus of example 1, wherein the frame is to hold the light spaced apart from the fan.

(45) Example 3 includes the apparatus of example 2, wherein the light is to be above the fan when the frame is mounted to the building structure via the articulated arm.

(46) Example 4 includes the apparatus of example 1, wherein the light and the fan are to point in substantially a same direction.

(47) Example 5 includes the apparatus of example 1, further including a controller to control operation of at least one of the light or the fan.

(48) Example 6 includes the apparatus of example 5, wherein the frame includes a rigid conduit extending between the light and the controller, the light to be electrically coupled to the controller via an electrical wire extending through the conduit.

(49) Example 7 includes the apparatus of example 5, wherein the frame supports the controller adjacent a back of the fan, the frame including a bend between the controller and the light, the bend

to extend around the fan to position the light proximate one of a top or a side of the fan.

(50) Example 8 includes the apparatus of example 1, wherein the fan is selectively mountable to the frame in a first orientation and a second orientation, the first orientation being rotated approximately 90 degrees relative to the second orientation, the first and second orientations corresponding to different placements of a discharge outlet of the fan relative to the frame.

(51) Example 9 includes the apparatus of example 8, wherein the fan is selectively mountable to the frame in a third orientation, the third orientation being rotated approximately 90 degrees relative to the first orientation, the third orientation being rotated approximately 180 degrees relative to the second orientation.

(52) Example 10 includes the apparatus of example 8, further include a frame plate attached to the frame, the frame plate including a symmetrical pattern of holes that align with a mounting surface of a housing of the fan in both the first orientation and the second orientation.

(53) Example 11 includes the apparatus of example 1, further including a handle to facilitate positioning of the light and the fan, the handle to be positioned below the light and below an axis of rotation of the fan.

(54) Example 12 includes the apparatus of example 11, wherein the handle includes a handlebar with two handgrips to extend substantially horizontally.

(55) Example 13 includes the apparatus of example 11, wherein the handle includes a single handgrip extending substantially vertically.

(56) Example 14 includes the apparatus of example 1, further including a mounting bracket to connect the articulated arm to the building structure via a pivotal connection, the pivotal connection defining an axis of rotation that is substantially perpendicular to an elongate length of the articulated arm.

(57) Example 15 includes the apparatus of example 14, wherein the mounting bracket includes a track and a track follower selectively moveable along the track, the articulated arm connected to the track follower, the track extending substantially parallel to the axis of rotation.

(58) Example 16 includes the apparatus of example 15, further including an actuator operatively coupled to at least one of the track or the track follower to enable selective movement of the track follower relative to the track.

(59) Example 17 includes the apparatus of example 1, further including the articulated arm, the articulated arm including a proximal arm to be pivotally coupled to the building structure, and a distal arm to be pivotally coupled to the proximal arm, the frame to pivot about both a first axis and a second axis relative to the distal arm, the first axis and the second axis being substantially perpendicular.

(60) Example 18 includes an apparatus comprising an illuminating fan assembly including a fan, a light, and a frame, the frame to support the fan and the light, and a positioning bracket to be connected to the frame at a first pivotal connection defining a first axis, the positioning bracket to be connect to an articulated arm at a second pivotal connection defining a second axis, the articulated arm to be mounted to a building structure adjacent a vehicle loading dock, the illuminating fan assembly to pivot, as a unit, about both the first axis and the second axis relative to the articulated arm, the first axis and the second axis being substantially perpendicular.

(61) Example 19 includes the apparatus of example 18, wherein the light is to be separated from the fan by the frame.

(62) Example 20 includes the apparatus of example 18, wherein the light and the fan are to point in substantially a same direction.

(63) Example 21 includes the apparatus of example 18, wherein the light is to be above the fan when the illuminating fan assembly is mounted to the building structure via the articulated arm.

(64) Example 22 includes the apparatus of example 18, further including a frame plate attached to the frame, the fan to be mounted to the frame via the frame plate.

(65) Example 23 includes the apparatus of example 22, wherein the frame plate enables the fan to

be selectively mounted to the frame in either a first orientation or a second orientation, the first orientation being rotated approximately 90 degrees relative to the second orientation, the first and second orientations corresponding to different placements of a discharge outlet of the fan relative to the frame.

(66) Example 24 includes the apparatus of example 23, wherein the frame plate enables the fan to be selectively mounted to the frame in a third orientation, the third orientation being rotated approximately 90 degrees relative to the first orientation and approximately 180 degrees relative to the second orientation.

(67) Example 25 includes the apparatus of example 22, wherein the fan is mounted to the frame plate via a mounting surface on a backside of a housing of the fan, the backside opposite a discharge outlet of the fan.

(68) Example 26 includes the apparatus of example 25, further including a controller to control operation of at least one of the light or the fan, the frame plate positioned between the controller and the fan.

(69) Example 27 includes the apparatus of example 26, wherein the frame includes a hollow bar extending between the light and the controller, the light to be electrically coupled to the controller via an electrical wire extending through the bar.

(70) Example 28 includes the apparatus of example 27, wherein the hollow bar includes a bend to extend around the fan to position the light above the fan with the controller behind the fan when the illuminating fan assembly is supported by the articulated arm.

(71) Example 29 includes the apparatus of example 18, further including a handle on the frame, the handle to be positioned lower than the light and lower than a midpoint of the fan.

(72) Example 30 includes the apparatus of example 29, wherein the handle includes a handlebar with two handgrips to extend substantially horizontally.

(73) Example 31 includes the apparatus of example 29, wherein the handle includes a single handgrip extending substantially vertically.

(74) Example 32 includes the apparatus of example 29, wherein the handle is an integral portion of the frame.

(75) Example 33 includes the apparatus of example 29, wherein the handle is a first handle, and further including a second handle spaced apart from the first handle, the second handle being an integral portion of the fan housing.

(76) Example 34 includes the apparatus of example 18, further including the articulated arm, and a mounting bracket to mount the articulated arm to the building structure, the mounting bracket including a track, the articulated arm moveable along the track to selectively adjust a position of the articulated arm relative to the mounting bracket.

(77) Example 35 includes the apparatus of example 34, further including an actuator operatively coupled to the mounting bracket to enable selective movement of the articulated arm along the track.

(78) Example 36 includes an apparatus for use at a vehicle loading dock, the apparatus comprising a frame to be pivotally coupled to an arm mountable to a building structure adjacent the vehicle loading dock, a fan to be coupled to the frame, a controller to be coupled to the frame, the controller to control operation of the fan, and a handle connected to the frame, the handle to be lower than the controller and lower than at least a midpoint of the fan when supported by the arm.

(79) Example 37 includes the apparatus of example 36, further including the arm, a mounting bracket having a track, the arm to be mounted to the building structure via the mounting bracket, and a track follower selectively moveable along the track, the arm to be pivotally coupled to the track follow such that as the track follower moves relative to the mounting bracket the arm moves with the track follower relative to the mounting bracket.

(80) Descriptors “first,” “second,” “third,” etc. are used herein when identifying multiple elements or components which may be referred to separately. Unless otherwise specified or understood



based on their context of use, such descriptors are not intended to impute any meaning of priority, physical order or arrangement in a list, or ordering in time but are merely used as labels for referring to multiple elements or components separately for ease of understanding the disclosed examples. In some examples, the descriptor “first” may be used to refer to an element in the detailed description, while the same element may be referred to in a claim with a different descriptor such as “second” or “third.” In such instances, it should be understood that such descriptors are used merely for ease of referencing multiple elements or components.

(81) As used herein, singular references (e.g., “a”, “an”, “first”, “second”, etc.) do not exclude a plurality. The term “a” or “an” entity, as used herein, refers to one or more of that entity. The terms “a” (or “an”), “one or more”, and “at least one” can be used interchangeably herein. Furthermore, although individually listed, a plurality of means, elements or method actions may be implemented by, e.g., a single unit or processor. Additionally, although individual features may be included in different examples or claims, these may possibly be combined, and the inclusion in different examples or claims does not imply that a combination of features is not feasible and/or advantageous.

(82) “Including” and “comprising” (and all forms and tenses thereof) are used herein to be open ended terms. Thus, whenever a claim employs any form of “include” or “comprise” (e.g., comprises, includes, comprising, including, having, etc.) as a preamble or within a claim recitation of any kind, it is to be understood that additional elements, terms, etc. may be present without falling outside the scope of the corresponding claim or recitation. As used herein, when the phrase “at least” is used as the transition term in, for example, a preamble of a claim, it is open-ended in the same manner as the term “comprising” and “including” are open ended. The term “and/or” when used, for example, in a form such as A, B, and/or C refers to any combination or subset of A, B, C such as (1) A alone, (2) B alone, (3) C alone, (4) A with B, (5) A with C, (6) B with C, and (7) A with B and with C. As used herein in the context of describing structures, components, items, objects and/or things, the phrase “at least one of A and B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, and (3) at least one A and at least one B. Similarly, as used herein in the context of describing structures, components, items, objects and/or things, the phrase “at least one of A or B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, and (3) at least one A and at least one B. As used herein in the context of describing the performance or execution of processes, instructions, actions, activities and/or steps, the phrase “at least one of A and B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, and (3) at least one A and at least one B. Similarly, as used herein in the context of describing the performance or execution of processes, instructions, actions, activities and/or steps, the phrase “at least one of A or B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, and (3) at least one A and at least one B.

(83) Although certain example methods, apparatus and articles of manufacture have been described herein, the scope of the coverage of this patent is not limited thereto. On the contrary, this patent covers all methods, apparatus and articles of manufacture fairly falling within the scope of the appended claims either literally or under the doctrine of equivalents.

## Claims

1. An apparatus comprising: a frame including an elongate bar, the bar to be pivotally coupled to a support system via a point of attachment on the bar between first and second ends of a longitudinal length of the bar; and a fan including an air outlet pointing in a first direction, the air outlet offset relative to a center of the fan, the fan selectively mountable to the frame in different orientations to change a location of the air outlet relative to the frame, the different orientations including a first orientation and a second orientation, the first orientation rotated relative to the second orientation

about an axis, the axis extending through the center of the fan, the axis generally parallel to the first direction.

2. The apparatus of claim 1, further including a frame plate attached to the frame, the frame plate including first holes that align with second holes in a mounting surface of a housing of the fan in both the first orientation and the second orientation.

3. The apparatus of claim 2, wherein the first and second holes in the frame plate are arranged in a symmetrical pattern.

4. The apparatus of claim 2, wherein the air outlet is located at a front of the fan, the mounting surface of the housing of the fan at a back of the fan opposite the front of the fan.

5. The apparatus of claim 2, further including a light, the bar to support the light such that the light remains in a same position relative to the frame regardless of whether the fan is in the first orientation or the second orientation.

6. The apparatus of claim 5, further including a controller to control operation of the light, the controller to be attached to the frame plate.

7. The apparatus of claim 6, wherein the frame plate is between the controller and the housing of the fan.

8. The apparatus of claim 1, wherein the air outlet is defined by a first dimension and a second dimension perpendicular to the first dimension, the second dimension greater than the first dimension, the air outlet offset relative to the center of the fan in a third direction parallel to the first dimension.

9. The apparatus of claim 8, wherein the second dimension corresponds to a width of the fan.

10. The apparatus of claim 8, wherein the second dimension is to extend in a fourth direction when the fan is in the first orientation and to extend in a fifth direction when the fan is in the second orientation, the fourth direction closer to a horizontal direction than to a vertical direction, the fifth direction closer to the vertical direction than to the horizontal direction.

11. The apparatus of claim 1, wherein the first orientation is rotated approximately 90 degrees relative to the second orientation.

12. The apparatus of claim 11, wherein the different orientations include a third orientation, the third orientation rotated approximately 90 degrees relative to the first orientation and rotated approximately 180 degrees relative to the second orientation.

13. The apparatus of claim 1, further including a handle connected to the frame, the handle to be lower than at least a midpoint of the fan when the fan is supported by the support system via the bar of the frame.

14. The apparatus of claim 13, wherein the handle is to remain at a fixed position relative to the frame regardless of whether the fan is in the first orientation or the second orientation.

15. The apparatus of claim 1, wherein the fan is to be positioned with the air outlet at a top of the fan when the fan is in the first orientation and at a lateral side of the fan when the fan is in the second orientation.

16. The apparatus of claim 1, wherein the center of the fan is defined by an axis of rotation of the fan.

17. An apparatus comprising: a bar pivotally couplable to an arm at a point of attachment on the bar between opposites ends of a longitudinal length of the bar; and a fan to be supported by the arm via the bar, the fan including a discharge outlet at a front of the fan, the discharge outlet defining a first area that is less than a second area corresponding to an outer perimeter of a projection of an entire body of the fan from the front of the fan, the discharge outlet spaced apart from a midpoint of the entire body of the fan, the fan selectively couplable to the bar in a first orientation or a second orientation, the first orientation rotated relative to the second orientation, the bar to extend above the fan when the fan is coupled to the bar in the first orientation, the bar to extend above the fan when the fan is coupled to the bar in the second orientation.

18. The apparatus of claim 17, further including a frame plate to attach the fan to the bar, the frame

plate to attach to a backside of the fan opposite the front of the fan.

19. An apparatus comprising: a fan; a frame including a bar, the frame to be supported by an arm pivotally couplable to the bar at a point of attachment between first and second ends of an elongate portion of the bar; and a handle to facilitate positioning of the fan, the handle to be vertically and horizontally offset relative to the point of attachment when the frame is supported by the arm, the elongate portion of the bar is to extend in a first direction, the handle to extend in a second direction different than the first direction, the fan to be attached to the bar in a fixed spatial relationship and fixed rotational relationship so that the positioning of the fan by the handle results in a corresponding movement of the bar, the positioning of the fan by the handle including movement of the fan relative to the arm and movement of the fan relative to a surrounding environment.

20. The apparatus of claim 19, further including a light, the bar to hold the light spaced apart from the fan, the light to be coupled to the bar at the first end, the fan to be coupled to the bar at the second end.

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